

Math 321: Groups and Symmetry

Personal revision notes I made for the 2025-2026 LU Math 321 exams

Chapter 0: Basics

- Common groups:
 - $C_n = \{1, w, w^2, \dots, w^{n-1}\}$ where $w = e^{2\pi i/n}$ together with multiplication
 - **Group of integers modulo n** is $\mathbb{Z}_n = \{\hat{0}, \hat{1}, \dots, \hat{n-1}\}$ together with addition modulo n.
 - **Symmetric group of order n** is S_n , the group of permutations of $\{1, 2, \dots, n\}$ together with composition
 - **Alternating group of order n** is A_n , the subgroup of S_n that consists of all the even permutations
 - **Dihedral Group** $D_{2n} = \{b^s a^t : b^2 = a^n = e, ab = ba^{n-1}\}$ together with multiplication
In D_{2n} , we have $a^k b = ba^{-k}$ for all $k \in \mathbb{Z}$
 - **Cyclic group** $\langle g \rangle$
If $C_n = \langle g \rangle$ where $g^n = e$ and $m|C_n$ then C_n has a subgroup of order m namely $H = \langle x^d \rangle$ where $d = n/m$
 - If G is a group and N is a normal subgroup of G , the **quotient group** G/N (read as "G modulo N"), is the set of all cosets of N in G , i.e., $G/N = \{gN : g \in G\}$ where the operation is defined as : $\forall g, h \in G, (gN)(hN) = (gh)N$

Chapter 1: Conjugacy

- Let G be a group. and $g, h \in G$. We say that g commutes with h if $gh = hg$.
- **Centraliser** The centraliser of g in G , $C_G(g)$, is the set of all elements in G which commute with g .
- How to calculate centralisers
 - Make a Cayley table and compare values

- **Centre** The centre of $Z(G)$ is defined as $Z(G) = \bigcap_{g \in G} C_G(g)$. It is the set of all elements in G which commutes with all the elements in G .

G is abelian iff $Z(G) = G$

$$Z(G) \triangleleft G$$

In a dihedral group, $G = D_{2n}$, $Z(G) = \{e\}$ when n is odd and $Z(G) = \{e, a^{n/2}\}$ when n is even.

- **Conjugate** If $g, h \in G$, h is conjugate to g if $(\exists x \in G)(h = xgx^{-1})$
Conjugacy is an equivalence relation on G .
- **Conjugacy Classes** The equivalence classes are called conjugacy classes.
 C_g is the conjugacy class containing g , i.e.,

$$C_g = \{h \in G : (\exists x \in G)(h = xgx^{-1})\}$$

$$C_g = \{g\} \iff g \in Z(G).$$

Properties preserved under conjugation: Order and size of centralisers

If g and h are conjugate and $h = xgx^{-1}$ then $C_G(h) = xC_G(g)x^{-1}$

$$|C_g| = |G|/|C_G(g)|$$

- Technique to find the conjugacy classes of a group:
 - Determine $Z(G)$. You can simply find elements which commute with every other element. Prove this!
 - Start with the elements in $Z(G)$. Each of these are in a class of its own.
 - Find elements which have the same order. They might be in the same class.
 - Check size of conjugacy class using $|C_g| = |G|/|C_G(g)|$. Bound it by finding a lower bound for $|C_G(g)|$
 - Check conjugacy conditions between elements and stop when we have enough elements to meet the size of the conjugacy class

A normal subgroup is a union of conjugacy classes of G .

- How to find the normal subgroups of a group
 - Find the conjugacy classes
 - Consider all the possible unions of these classes

- Check it is a subgroup using Lagrange's Theorem.
- If you multiply any two elements in your union, the result must stay inside the union.
Remember: Any subgroup must contain the identity!! So the union must always contain C_e

- **Cycle Type** Let $\pi \in S_n$. The cycle type $\text{type}(\pi)$ of π is the list of lengths of cycles in the full disjoint cycle notation for π in order of decreasing size.

Cycle type is preserved under conjugation

- Two permutations are conjugate iff they have the same cycle type.
Remember: when comparing cycle types, make sure both are written in disjoint cycle notation!!
- For conjugate elements of S_n , π and ρ , how to find σ such that $\rho = \sigma\pi\sigma^{-1}$
 - Write ρ below π ensuring the cycles of the same length are matched against each other. Let σ be the permutation sending each element of the first row to the element below it.
- How do you calculate the size of the conjugacy class C_π given $\pi \in S_n$?
 - We need to count the number of permutations having the same cycle type as π
 - Suppose π has b_i cycles of length i . Then

$$|C_\pi| = \frac{n!}{1^{b_1} 2^{b_2} \dots n^{b_n} b_1! b_2! \dots b_n!}$$

Chapter 2: Direct Products

- **External Direct Product** If G_1 and G_2 are groups, the external direct product is the group $G_1 \times G_2$ with the binary operation $(g_1, g_2)(h_1, h_2) = (g_1 h_1, g_2 h_2)$. G_1 and G_2 are called **factors** of $G_1 \times G_2$
- A direct product is abelian iff its factors are abelian
- $o(g_1, g_2) = \text{lcm}(o(g_1), o(g_2))$
- r and s are coprime iff $C_r \times C_s \cong C_{rs}$

- Checking if a direct product is isomorphic to some group G :
 - Compare the orders of the elements in the direct product and the orders of the elements in G
 - Check if both G and the direct product is abelian
- Finding subgroups of a Direct Product
 - If $H_1 \leq G_1$ and $H_2 \leq G_2$ then $H_1 \times H_2 \leq G_1 \times G_2$.
 Note: The direct product can have subgroups that is not a direct product of subgroups of the factors
- Criterion for a group to be a direct product, i.e., what conditions G must hold for there to be an isomorphism $\theta : G \rightarrow G_1 \times G_2$ such that $\theta(G_i) = \tilde{G}_i$?
 - G must have normal subgroups G_1 and G_2 such that $G = G_1 G_2$ and $G_1 \cap G_2 = \{e\}$ (i.e., every element of g can be written uniquely as $g = g_1 g_2$ with $g_i \in G_i$).
- **Internal Direct Product** If G has normal subgroups G_1 and G_2 satisfying the criterion, then we call G the internal direct product of G_1 and G_2 and write $G = G_1 \times G_2$.
- Finite abelian groups classification Every finite abelian group is isomorphic to a direct product of cyclic groups in the form

$$C_{p_1^{r_1}} \times C_{p_2^{r_2}} \times \cdots \times C_{p_t^{r_t}}$$

where the p_i are primes (not necessarily distinct) and the product is unique up to reordering of the factors

- How to find all abelian groups of order n (up to isomorphism)
 - Factorise n into a product of prime powers
 - Find all possible sets of prime powers
 - If $\{p_1^{r_1}, p_2^{r_2}, \dots, p_t^{r_t}\}$ is a set of prime power then

$$C_{p_1^{r_1}} \times C_{p_2^{r_2}} \times \cdots \times C_{p_t^{r_t}}$$

is one of the groups of order n .

- Standard way of writing a finite abelian group G as a direct product of cyclic groups:
 - For each distinct prime dividing $|G|$, write the powers which appear in a row, in ascending order of size, keeping the right hand ends of the various rows aligned. Each prime having its own row.
 - Multiply together the numbers occurring withing each column to form numbers $m_1, m_2, \dots, m_l$.
 - Then $G \cong C_{m_1} \times C_{m_2} \times \dots \times C_{m_l}$ and $m_i | m_{i+1}$ for all $i \in \{1, \dots, l-1\}$.
- Classification of subgroups of finite abelian groups

Finite abelian groups have subgroups of all orders allowed by Lagrange's Theorem, i.e., if G is abelian with order n and $m | n$ then G has a subgroup of order m .
- How to find subgroups of finite abelian groups
 - Write $G = C_{p_1^{r_1}} \times C_{p_2^{r_2}} \times \dots \times C_{p_t^{r_t}}$
 - To find a subgroup of order m , since $m | n$ we write

$$m = p_1^{t_1} p_2^{t_2} \dots p_t^{t_t}$$
 - for each i , write $C_{p_i^{r_i}} = \langle x \rangle$ and $p_i^{r_i} = d p_i^{t_i}$. Then $H_i = \langle x^d \rangle$ is a subgroup of $C_{p_i^{r_i}}$
 - $H_1 \times H_2 \times \dots \times H_t$ is a subgroup of G .
- Condition which forces an abelian group to be cyclic If an abelian group has order n and n is a product of distinct primes then G is cyclic.

Chapter 3: Jordan- Hölder Theorem

- Facts about normal subgroups of a finite group G
 - G abelian $\implies$ all subgroups are normal
 - $\{e\}$ and G are normal
 - a subgroup with index 2 is normal
 - a subgroup which is the union of conjugacy classes is normal
 - $Z(G)$ is normal

– Kernel of a homomorphism is normal

- **First Isomorphism Theorem**

If $\phi : G \rightarrow H$ is a homomorphism then $G/\ker \phi \cong \text{im} \phi$.

- **Second Isomorphism Theorem**

If $N \triangleleft G$ and $H \leq G$ then $(NH)/N \cong H/(N \cap H)$

- **Simple group** A non-trivial group G is simple iff it has no proper non-trivial normal subgroups.

- Classification of simple abelian groups

Suppose G is abelian. Then G is simple iff $G \cong C_p$ for some prime p .

- **Maximal** A proper normal subgroup $N \triangleleft G$ is called maximal if N is not contained in any normal subgroup of G except G and itself, i.e, If M is normal and $N \triangleleft M$ then $M = N$ or $M = G$.

- A normal subgroup $N \triangleleft G$ is maximal iff G/N is simple.

- If $N \triangleleft G$ satisfies $|G|/|N| = p$ for some prime p then N is maximal and $G/N \cong C_p$

- **Composition series** A composition series for G is a series of subgroups

$$G = G_0 \geq G_1 \geq G_2 \geq \cdots \geq G_r = \{e\}$$

such that $G_i \triangleleft G_{i-1}$ and G_{i-1}/G_i is simple for all i . The simple groups G_{i-1}/G_i are called the **composition factors** of G .

- How to find the composition factor of a group

– Start with a non-trivial group G and choose a maximal normal subgroup, call it G_1 . (The trivial subgroup is such a group so a maximal one exists.) You can check G_1 is maximal by calculating $|G|/|G_1|$.

– Repeat the procedure and choose a maximal normal subgroup of G_i for $i \geq 1$. until we obtain the trivial subgroup

- **Jordan- Hölder Theorem** The list of composition factors for any two composition series is the same

We can associate a collection of simple groups (the composition factors) to a given finite group

- **Extension of N by Q** If G, N , and Q are groups with $N \trianglelefteq G$ and $G/N \cong Q$ then G is called an extension of N by Q .
- **Soluble** A group G is soluble if its composition factors are all abelian
 G abelian $\implies G$ soluble.
- Sufficient conditions for $N \triangleleft A_n$ to be equal to A_n where $n \geq 5$, i.e., for N to be a simple group.
 - If N contains a 3-cycle
 - If N contains a product of two disjoint transpositions
- How to find a 3-cycle in $N \triangleleft A_n$ given $\pi \in N$
 - Write π as a product of disjoint cycles $\pi = \gamma_1 \gamma_2 \cdots \gamma_k$ in non-increasing length
 - Case 1: $\gamma_1 = (a_1 a_2 \cdots a_m)$ where $m > 3$. Set $\sigma = (a_1 a_2 a_3)$ and calculate $\sigma \pi \sigma^{-1} \pi^{-1}$. You should get $(a_1 a_2 a_4)$
 - Case 2: $\gamma_1 = (a_1 a_2 a_3)$ and $\gamma_2 = (a_4 a_5 a_6)$. Set $\sigma = (a_2 a_3 a_4)$ and calculate $\sigma \pi \sigma^{-1} \pi^{-1}$. You should get $(a_1 a_2 a_4 a_3 a_5)$. Apply case 1.
 - Case 3: $\gamma_1 = (a_1 a_2 a_3)$ and $\gamma_2 \gamma_3 \cdots \gamma_k$ are transpositions. Calculate π^2 . You should get $(a_1 a_3 a_2)$
 - Case 4: $\gamma_1 \gamma_2 \cdots \gamma_k$ are all transpositions. Let $\gamma_1 = (a_1 a_2)$ and $\gamma_2 = (a_3 a_4)$. Set $\sigma = (a_2 a_3 a_4)$. Calculate $\sigma \pi \sigma^{-1} \pi^{-1}$. You should get $(a_1 a_4)(a_2 a_3)$
- Conclusion: A_n where $n \geq 5$. is an example of a non-abelian simple group and hence non-soluble group.

Chapter 4: Group Actions

- **Action** We say that a group G acts on a non-empty set X if for all $g \in G$ and for all $x \in X$ there exists a unique $gx \in X$ such that

$$(g_1 g_2)x = g_1(g_2 x) \\ ex = x$$

Note this is a function $G \times X \rightarrow X$.

- Examples of actions
 - **Rotation** Let $G = \mathbb{R}$ (under addition) and $X = \mathbb{R}^2$. Define action $G \times X \rightarrow X$ by $g(x, y) \mapsto R_g(x, y)$
 - **Natural Action** Let $G = S_n$ and $X = \{1, 2, \dots, n\}$. Define the action $G \times X \rightarrow X$ by $\pi i \mapsto \pi(i)$ for all $\pi \in S_n$ and $i \in X$.
- **Orbit** If G acts on X , the orbit of $x \in X$ under G is the set $O_x = \{gx : g \in G\}$. The set of points to which x can be moved by elements of G .
- If G acts on X , the orbits of the action partition X .
- **Stabiliser** If G acts on X , the stabiliser of $x \in X$ are all the elements in G which fixes x i.e.,

$$\text{stab}_G(x) = \{g \in G : gx = x\}$$

- Stabilisers are subgroups of G
- **Orbit-Stabiliser Theorem** If G acts on X and $x \in X$ then

$$|O_x| = |G : \text{stab}_G(x)|$$

If G is finite, by Lagrange's Theorem $|O_x||\text{stab}_G(x)| = |G|$.

- Points lying in the same orbit have conjugate stabilisers, i.e.,

$$gx = y \implies g\text{stab}_G(x)g^{-1} = \text{stab}_G(y)$$

- **Polyhedron** Solid shape whose faces are polygons. It is **regular** if all the faces are regular polygons of the same size and shape with the configuration at each vertex being the same.
- Five types of regular polyhedron

Chapter 5: p -groups and p -subgroups

- **p -group** A group whose order is a positive power of p where p is prime
- **p -subgroup** A subgroup of G whose order is a power of p
- **Sylow's First Theorem** If G is a group of order $p^m r$ where $p \nmid r$, then G has a subgroup of order p^m

- **Sylow p -subgroup** (*Tells us a sufficient condition for a group to have a p -subgroup*)

If G is a group of order $p^m r$ where $p \nmid r$, then a subgroup of order p^m is called a Sylow p -subgroup of G .

- **Sylow's Second Theorem** (*Relates the conjugacy of two Sylow p -subgroups*)

If G is a finite group, then any two Sylow p -subgroups P_1, P_2 of G are conjugate in G , i.e., there exists $g \in G$ such that $P_2 = gP_1g^{-1}$.

- **Sylow's Third Theorem** (*gives us information on the number of Sylow p -subgroups*)

If G is a group of order $p^m r$ where $p \nmid r$ and n_p is the number of distinct Sylow p -subgroups of G , then $n_p \mid r$ and $n_p \equiv 1 \pmod{p}$, i.e., $p \mid (n_p - 1)$